

Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



Project Overview

Goal

Analyze 3,900 purchases to understand spending patterns, segments, and preferences.

Objective

Optimize operations, reduce costs, and improve forecasting.



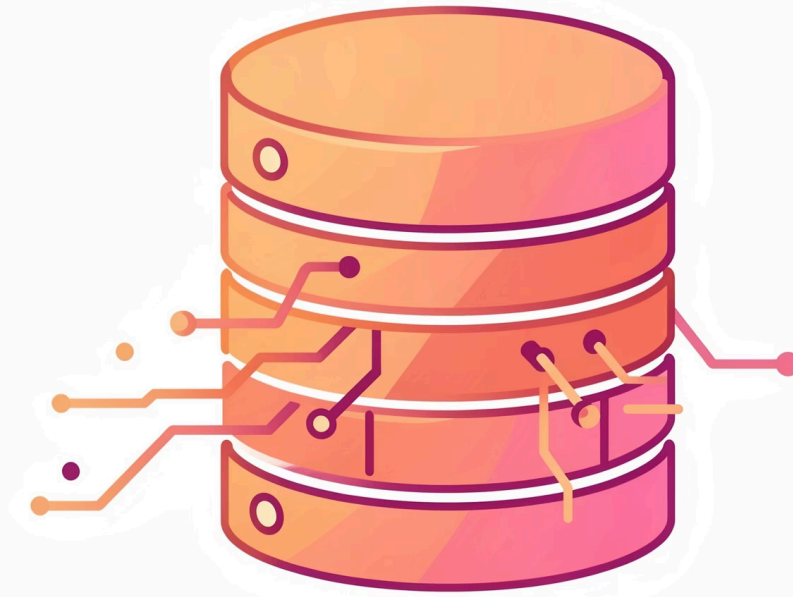
Dataset Summary

Rows: 3,900

Columns: 18

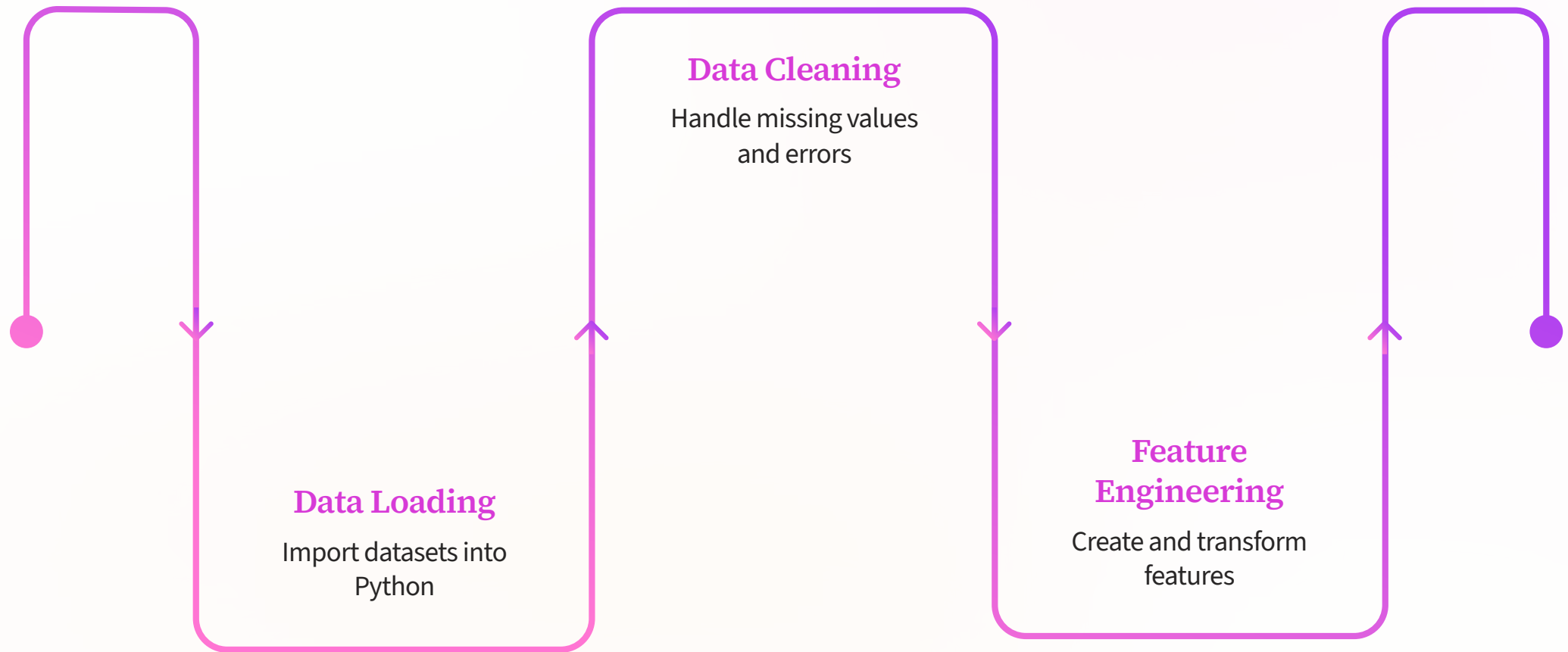
Key Features:

- Customer demographics (Age, Gender, Location, Subscription)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)



Missing Data: 37 values in Review Rating column.

Exploratory Data Analysis (Python)



Initial data preparation and cleaning steps performed using Python for robust analysis.

Data Analysis (SQL)



Revenue by Gender

Male: \$157,890 | Female:
\$75,191



High-Spending Discount Users

Identified customers using
discounts, spending above
average.



Top 5 Products by Rating

Gloves (3.86), Sandals (3.84),
Boots (3.82), Hat (3.80), Skirt
(3.78)

SQL Analysis: Shipping & Subscriptions

Shipping Type Comparison

Average purchase amounts:

- Standard: \$58.46
- Express: \$60.48



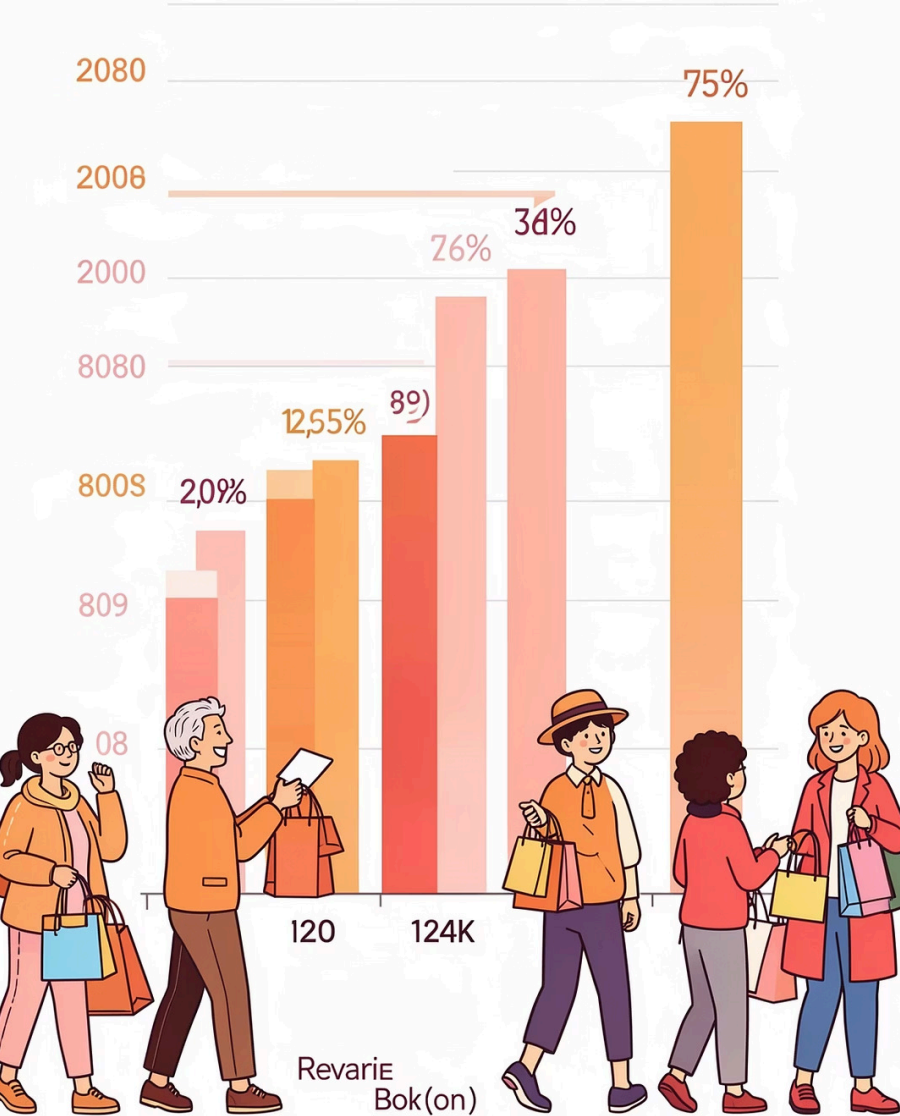
Subscribers vs. Non-Subscribers

Average spend and total revenue:

- Subscribers: Avg \$59.49, Total \$62,645
- Non-Subscribers: Avg \$59.87, Total \$170,436



BE REVENUE BY AGE GROUP



SQL Analysis: Product & Age Insights

Top 3 Products per Category

Most purchased items within Accessories, Clothing, Footwear, Outerwear.

Repeat Buyers & Subscriptions

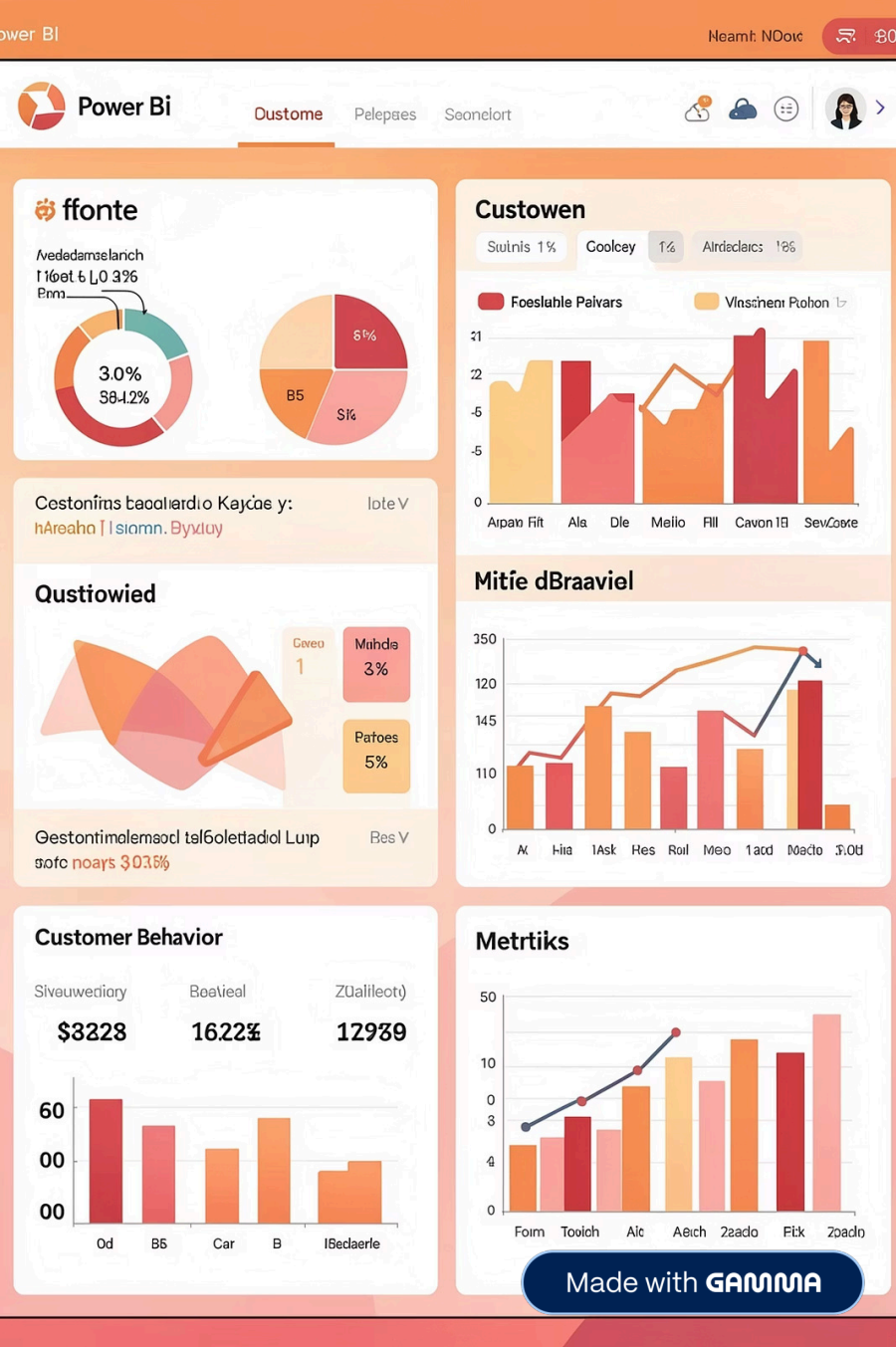
Customers with >5 purchases: 958 subscribed, 2518 not subscribed.

Revenue by Age Group

Young Adult: \$62,143 | Middle-aged: \$59,197 | Adult: \$55,978 | Senior: \$55,763

Interactive Dashboard in Power BI

Visualizing key insights for easy interpretation and strategic decision-making.





Key Dashboard Metrics

3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Review

Business Recommendations

→ **Boost Subscriptions**

Promote exclusive benefits for subscribers.

→ **Customer Loyalty Programs**

Reward repeat buyers to foster loyalty.

→ **Review Discount Policy**

Balance sales boosts with margin control.

→ **Product Positioning**

Highlight top-rated and best-selling products.

→ **Targeted Marketing**

Focus on high-revenue age groups and express-shipping users.